Expected Experiment Matrix

Exact table structure of the current paper, with expected values

v02 · 2026-10-09 · planning targets only. No cell is an observed result, confidence interval, or closure decision.

Binding. The row order, column semantics, experiment IDs, and displayed methods follow the canonical `main.tex` and its 9-page `main.pdf`. Exact hashes are in `main_input_sha256.txt`. The preview is not the binding table numbering.

Reading convention. Numbers are conservative targets anchored to the current recorded table and its documented estimands. They express the intended direction: natural stationary routing is tied to or slightly below Frozen; controlled correctness drift is where MIRA-Relay should outperform Frozen and the matched Online-SGD row; the development objectives remain below Corrected Frozen; and cost is a separate trade-off. A target range is a planning band, never evidence.

Table 1 — Comparators under the common routing contract

Role	Expected displayed methods
Anchors	Random; fit-only Best Fixed; retrospective Best Fixed[†]
Matched updates	Frozen; MIRA-Relative; Online SGD
Online controls	Greedy LinUCB; BayesianRouter
Proposed	MIRA-Relay
Mechanism	Fixed cyclic permutation credit
Audit only	Pooled-feedback static recentering

[†] Evaluator-only headroom reference. The cyclic control is a sealed replay; pooled recentering is excluded from mechanism claims. `Online SGD` is the displayed label for the matched method ID `matched_online_sgd`.

Table 2 — Experimental map

ID	Question	Expected evidence location
E1	Adaptation and drift	Table 3 and Figure 3
E2	Source-aligned credit	Tables 4 and 3
E3	Initial selection	Table 5
E4	Scale and centering	Table 6
E5	Policy cost / validity	Table 8
E6	Post-call refusal	Table 7

This map is copied as a structural table; it contains no acceptance thresholds.

Table 3 — Selected correctness (%)

Original order and the mean over 128 uniform random orders of the same previously inspected cohort. The controlled columns use the declared evaluator-side correctness drift.

	Natural		Cont
Method	Original	128-order mean	Original
Random	83.8	86.5	—
Best Fixed	88.2	88.2	—
Frozen	89.8	89.8	73.4
MIRA-Relative	89.8	89.8	—
Online SGD	89.2	89.1	84.0
Greedy LinUCB	89.7	89.7	81.2
BayesianRouter	89.5	89.7	80.3
MIRA-Relay	**89.8**	89.5	**87.6**
Fixed-permutation credit	—	89.9	—
Retrospective Best Fixed†	90.5	90.5	—

The MIRA-Relative natural target equals Frozen because its molecule-independent multiplicative factor leaves the action sequence unchanged. The natural MIRA target is a null/slight-loss pattern; its controlled target is below any evaluator-only ceiling and above the matched online controls. † Evaluator-only.

Figure 3 data — expected gain over Frozen (pp)

Panel	Method	$D = 0$	$D = 1$	$D = 4$
Natural	Online SGD	−0.76	−0.89	−0.13
Natural	Greedy LinUCB	−0.13	0.00	−0.13
Natural	BayesianRouter	−0.25	−0.25	−0.32
Natural	MIRA-Relay	−0.13	−0.13	+0.25
Controlled drift	Online SGD	+11.59	+11.21	+9.17
Controlled drift	Greedy LinUCB	+8.66	+7.01	+7.64
Controlled drift	BayesianRouter	+6.82	+6.94	+6.94
Controlled drift	MIRA-Relay	+13.63	+14.52	+14.39

These are the points plotted in Figure 3, not a second estimand. The controlled MIRA and Online-SGD means are approximately +14.18 and +10.66 pp, matching the intended Table 3 original-order contrast. No uncertainty band is implied.

Table 4 — Source-aligned attribution

Contrast	Natural Δ [planning band]	Controlled Δ [planning band]
Local − Frozen	+0.10 [−1.08, +1.42]	+14.40 [+9.79, +19.23]
Permutation − Frozen	+0.41 [−0.47, +1.32]	−1.27 [−3.66, +0.95]
Local − Permutation	−0.30 [−1.42, +0.76]	+15.67 [+10.65, +20.74]

Units: pp. Equal-weight scaffold-mean contrasts over 124 clusters and 90 replay streams per condition. The bands are planning ranges, not observed confidence intervals; this estimand is separate from Table 3 pooled correctness.

Table 5 — Cold-start accuracy on BBB development

Selector	Accuracy (%)
Best Fixed	79.2
Raw Frozen	79.4
Corrected Frozen	**80.2**
BCE + auxiliary CLS	78.5 / 79.0
BT + auxiliary CLS	78.8 / 78.3
Per-molecule oracle†	85.6

The two values for each learned objective are paired training seeds 0/1. Both objectives are expected to remain below Corrected Frozen, preserving the documented negative diagnostic. † Uses unselected outputs and is evaluator-only.

Table 6 — Score scale and update rule

Score	Update	Natural	Drift
Raw	Frozen	79.4	69.1
	Marginal	79.2	77.3
	Query	78.5	76.0
Corrected	Frozen	80.2	63.3
	Marginal	79.8	78.3
	Query	**80.3**	**78.7**

This is a development diagnostic. Query-relative updating is expected to improve drift modestly; its natural effect remains small and does not establish a main-method gain.

Table 7 — Post-call refusal

Policy	Coverage (%)	Accepted accuracy (%)	Utility
Always accept	100.00	88.15	−0.1285
Confidence refuse	84.46	**94.57**	**−0.0947**
Always refuse	0.00	—	−0.2600
Same-coverage random†	84.46	88.69	−0.1444

Secondary analysis on the reused holdout. Expected integer tuples (W, R, C, N) are $(93, 0, 785, 785)$ for always accept, $(36, 122, 785, 785)$ for confidence refuse, $(0, 785, 0, 785)$ for always refuse, and $(75, 122, 785, 785)$ for same-coverage random. Coverage and utility use these counts and $U = -(W + 0.25R + 0.01C)/N$. † Evaluator-side conditional null.

Table 8 — CPU policy overhead (μs)

Expected mean of 30 stream-level p95 timings on macOS arm64. Provider inference, I/O, service latency, and live memory are separate quanti-

	Natural menu		Controlled
Method	Selection	Update	Update
Random	67.1	5.4	—
Best Fixed	5.8	0.7	—
Frozen	15.9	1.3	1.5
MIRA-Relative	42.0	73.3	—
Online SGD	14.1	19.4	20.0
Greedy LinUCB	128.6	90.9	92.3
BayesianRouter	133.7	89.0	92.6
MIRA-Relay	**27.2**	**2.8**	**3.5**

ties.

Trade-off target: MIRA-Relay costs more selection time than Frozen but less than LinUCB/BayesianRouter, while its update remains lightweight. This table does not assert a joint p95 cap or end-to-end service cost.

References. Canonical schema: `main.tex`; E1–E6 semantics: `experiment_tables/README.md`. Baseline anchors: Li et al. (2010) and Agrawal and Goyal (2013). The Guide v09 audits are review inputs only; its status remains HOLD.